MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
DSE – MG4DSEMAT201 – TRANSFORMS AND FOURIER SERIES
Scheme of Valuation

Duration: 2 Hours **Maximum Marks: 70**

Part A

Short answer type Questions
Answer any 5 questions
Each question carries 2 marks

1. Correct formula: $\mathcal{L}\{u(t-a)\} = \dfrac{e^{-as}}{s}$ ..... 2 marks [U]/[CO1]

2. Property used: $\mathcal{L}\{\int_0^t f(\tau)d\tau\} = \dfrac{1}{s}F(s)$ ..... 1 mark [U]/[CO1]

   Final answer: $\dfrac{1}{s(s^2+1)}$ ..... 1 mark

3. Splitting: $\dfrac{s+1}{s^2} = \dfrac{1}{s} + \dfrac{1}{s^2}$ ..... 1 mark [A]/[CO2]

   Inverse + shifting: $tu(t-1)$ ..... 1 mark

4. Convolution property ..... 1 mark [U]/[CO2]

   Final: $\dfrac{1}{(s-a)(s-b)}$ ..... 1 mark

5. Fourier cosine series: $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$ ..... 1 mark [K]/[CO3]

   $a_0 = \frac{1}{L}\int_0^L f(x)\,dx$, $a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$ ..... 1 mark [K]/[CO3]

6. Even periodic extension is defined by $f(-x) = f(x)$ and the extended function has period $2L$ ..... 1 mark [K]/[CO3]

   OR .....

   Using $a_0 = \frac{1}{L}\int_0^L f(x)\,dx$ and $a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$, we obtain Fourier cosine series (even periodic extension) ..... 2 marks [K]/[CO3]

7. $\mathcal{F}\{f''(x)\} = -\omega^2\mathcal{F}\{f(x)\}$ ..... 2 marks [K]/[CO4]

8. Piecewise continuity ..... 1 mark [U]/[CO4]

   Absolute integrability ..... 1 mark

Part B

Short Essay Type Questions
Answer any 5 questions
Each question carries 6 marks

9. $\mathcal{L}\{\sinh at\} = \dfrac{a}{s^2 - a^2}$ ..... 1 mark [A]/[CO1]

$\mathcal{L}\{tf(t)\} = -\dfrac{d}{ds}F(s)$ ..... 1 mark

$\mathcal{L}\{t\sinh at\} = -\dfrac{d}{ds}\left(\dfrac{a}{s^2 - a^2}\right)$ ..... 2 marks

$= \dfrac{2as}{(s^2 - a^2)^2}$ ..... 2 marks [A]/[CO1]

10. $\dfrac{4s+1}{s^2-16} = \dfrac{4s}{s^2-16} + \dfrac{1}{s^2-16}$ ..... 2 marks [U]/[CO1]

$\mathcal{L}^{-1}\left\{\dfrac{4s}{s^2-16}\right\} = 4\cosh 4t$ ..... 2 marks

$\mathcal{L}^{-1}\left\{\dfrac{1}{s^2-16}\right\} = \dfrac{1}{4}\sinh 4t$ ..... 1 mark

Answer $= 4\cosh 4t + \dfrac{1}{4}\sinh 4t$ ..... 1 mark

11. $\cot^{-1}\dfrac{s}{a} = \tan^{-1}\dfrac{a}{s}$ ..... 2 marks [A]/[CO2]

$\mathcal{L}^{-1}\left\{\tan^{-1}\dfrac{a}{s}\right\} = \dfrac{\sin at}{t}$ ..... 2 marks

Therefore, $\mathcal{L}^{-1}\left\{\cot^{-1}\dfrac{s}{a}\right\} = \dfrac{\sin at}{t}$ ..... 2 marks

12. $\dfrac{5s}{s(s+3i)(s+4i)} = \dfrac{5}{(s+3i)(s+4i)}$ ..... 1 mark [A]/[CO2]

$\dfrac{5}{(s+3i)(s+4i)} = \dfrac{-5i}{s+3i} + \dfrac{5i}{s+4i}$ ..... 3 marks

Taking inverse Laplace transform gives $-5ie^{-3it} + 5ie^{-4it}$ ..... 1 mark

Answer $= 5i(e^{-4it} - e^{-3it})$ ..... 1 mark

13. $a_0 = 20, \quad a_n = 0$ for $n \geq 1$ ..... 2 marks [U]/[CO3]

$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$ ..... 1 mark

$b_n = \dfrac{14(-1)^{n+1}}{n}$ ..... 2 marks

$f(x) = 10 + 14\left(\sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \cdots\right)$ ..... 1 mark

14. Identifying odd function .....1 mark [U]/[CO3]

Formula for b_n .....2 marks

Integration .....2 marks

Final series .....1 mark

15. $F(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)e^{-i\omega x}\,dx$ ..... 1 mark [A]/[CO4]

$= \frac{1}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-kx}e^{-i\omega x}\,dx$ .....2 marks

$= \frac{1}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-(k+i\omega)x}\,dx$ .....1 mark

Evaluating integral, $F(\omega) = \frac{1}{\sqrt{2\pi}(k+i\omega)}$ .....2 marks

16. $F(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)e^{-i\omega x}\,dx$ ..... 1 mark [A]/[CO4]

Given $f(x) = kx,\ a < x < b,\ 0$ otherwise .....1 mark

$F(\omega) = \frac{k}{\sqrt{2\pi}} \int_{a}^{b} xe^{-i\omega x}\,dx$ .....2 marks

Using integration by parts .....1 mark

$F(\omega) = \frac{k}{\sqrt{2\pi}} \left[\frac{e^{-i\omega b}(1-i\omega b) - e^{-i\omega a}(1-i\omega a)}{\omega^2}\right]$ .....1 mark

Part C

Essay Type Questions
Answer any 3 questions
Each question carries 10 marks

17. **(a)** Using linearity, $\mathcal{L}\{4e^{5t} + 6t^3 - 3\sin 4t\} = 4\mathcal{L}\{e^{5t}\} + 6\mathcal{L}\{t^3\} - 3\mathcal{L}\{\sin 4t\}$ ....2 marks [U]/[CO1]

Using $\mathcal{L}\{e^{5t}\} = \frac{1}{s-5}$, $\mathcal{L}\{t^3\} = \frac{3!}{s^4}$, $\mathcal{L}\{\sin 4t\} = \frac{4}{s^2+16}$ ..... 1 mark

$= \frac{4}{s-5} + \frac{36}{s^4} - \frac{12}{s^2+16}$ .....2 marks

(b) $\mathcal{L}\{t^4 - \sinh 2t + 8\} = \mathcal{L}\{t^4\} - \mathcal{L}\{\sinh 2t\} + \mathcal{L}\{8\}$ .....2 marks

$= \frac{24}{s^5} - \frac{2}{s^2-4} + \frac{8}{s}$ .....3 marks

18. $\mathcal{L}^{-1}\left\{\ln\left(\dfrac{s+1}{s}\right)\right\} = \dfrac{1-e^{-t}}{t}$ ..... 5 marks [A]/[CO2]

$\mathcal{L}^{-1}\left\{\tan^{-1}\left(\dfrac{2}{s}\right)\right\} = \dfrac{\sin 2t}{t}$ ..... 5 marks

19. **(a)** $\dfrac{3-2s}{(s+1)(s+2)(s+6)} = \dfrac{A}{s+1} + \dfrac{B}{s+2} + \dfrac{C}{s+6}$ ..... 2 marks [A]/[CO2]

$A = -1, \quad B = \dfrac{7}{4}, \quad C = -\dfrac{3}{4}$ .....2 marks

Answer $= -e^{-t} + \dfrac{7}{4}e^{-2t} - \dfrac{3}{4}e^{-6t}$ .....1 mark

(b) $\dfrac{6s}{(s+1)^2(s-2)^2} = \dfrac{A}{s+1} + \dfrac{B}{(s+1)^2} + \dfrac{C}{s-2} + \dfrac{D}{(s-2)^2}$ .....2 marks

$A = -\dfrac{2}{3}, \quad B = -\dfrac{2}{3}, \quad C = \dfrac{2}{3}, \quad D = \dfrac{4}{3}$ ..... 2 marks

Answer $= -\dfrac{2}{3}e^{-t} - \dfrac{2}{3}te^{-t} + \dfrac{2}{3}e^{2t} + \dfrac{4}{3}te^{2t}$ .....1 mark

20. Given function is odd $\Rightarrow a_0 = 0,\ a_n = 0$ ..... 2 marks [U]/[CO3]

Only sine coefficients required .....1 mark

$b_n = 0$ for even $n, \quad b_n = \dfrac{4}{n\pi}$ for odd n .....3 marks

Writing general Fourier series expansion .....1 mark

$f(x) = \sum_{n=1,3,5,\ldots}^{\infty} \dfrac{4}{n\pi}\sin(n\pi x)$ .....2 marks

$= \dfrac{4}{\pi}\left(\sin\pi x + \dfrac{1}{3}\sin 3\pi x + \dfrac{1}{5}\sin 5\pi x + \cdots\right)$ ..... 1 mark

21. Given $f(x) = k,\ 0 < x < a$ and $f(x) = 0,\ x > a$ .....1 mark [A]/[CO4]

Fourier cosine transform: $\hat{f}_c(\omega) = \sqrt{\dfrac{2}{\pi}}\int_0^{\infty} f(x)\cos\omega x\,dx$ ..... 2 marks

$= \sqrt{\dfrac{2}{\pi}}\,k\int_0^{a}\cos\omega x\,dx$ ..... 1 mark

$= \sqrt{\dfrac{2}{\pi}}\,k\left(\dfrac{\sin a\omega}{\omega}\right)$ ..... 2 marks

Fourier sine transform: $\hat{f}_s(\omega) = \sqrt{\dfrac{2}{\pi}}\int_0^{\infty} f(x)\sin\omega x\,dx$ .....2 marks

$= \sqrt{\dfrac{2}{\pi}}\,k\int_0^{a}\sin\omega x\,dx$ ..... 1 mark

$= \sqrt{\frac{2}{\pi}}\, k \left(\frac{1 - \cos a\omega}{\omega} \right)$ ..... 1 mark

22. Definition of convolution: $(f * g)(x) = \int_{-\infty}^{\infty} f(p)\, g(x-p)\, dp$ ..3 marks [U]/[CO4]

Fourier transform: $\mathcal{F}\{f * g\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(p)\, g(x-p) e^{-i\omega x}\, dp\, dx$ 2 marks

Interchange order of integration ..... 1 mark

Substitute $x - p = q$ ..... 1 mark

Split into product of two integrals ..... 1 mark

$\mathcal{F}\{f * g\} = \sqrt{2\pi}\, \mathcal{F}\{f\}\mathcal{F}\{g\}$ ..... 2 marks